

Raspberry Pi remotely, PHP can easily be installed by the following command in the terminal:

```
$ sudo apt-get install php5
```

This command runs without any issue and eventually comes back to the prompt. Once completed, install the PHP-GPIO libraries to interact with the GPIOs. Download the libraries from 'https://github.com/aaronpearce/PHP-GPIO.'

Extract the libraries and paste them into '/var/www' (take care to only paste the files, not the folder itself).

Download the source code 'stepper4.php' for four-step sequence and 'stepper8.php' for eight-step sequence and copy it to '/var/www'. Now use 'cd' command to go to the directory where the source code is located and use the command below to run the software:

```
$ sudo php -f stepper4.php
```

where 'stepper4.php' is the name of the software.

You should see the motor rotating now.

Project 2: Fine control with stepper motor

This project solves a real-time problem in the industry where speed control in various old machines is still done with a mechanical lever. It can be highly useful if a precise remote speed control is possible. It takes too much time to make speed adjustments manually, which affects production.

Fine speed control can easily be done by a powerful stepper motor controlling the lever. The control command can be sent to Raspberry Pi remotely, which further controls stepper motor precisely. Servo control, which could be another alternative, is ruled out due to its plastic body which

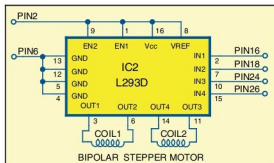


Fig. 3: Circuit for fine control with stepper motor

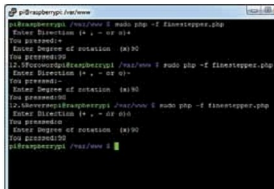


Fig. 4: Output of the program showing various options

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on www.efymag.com website.

doesn't suit harsh environmental conditions.

Circuit and working

Fig. 3 shows the circuit for fine control with stepper motor. Here we have used dual H-bridge L293D (IC2) driver, which can deliver 0.6A current. Bipolar stepper motors also have two coils like unipolar stepper motors but without the centre taps.

The stepper motor we used here runs on 5 V. Therefore we just connected pin 8 of IC2 to pin 2 of Raspberry

Pi, which has on board 5 V. But in case more power is drawn than that could be supplied by the Raspberry Pi board, the Raspberry Pi may reboot. In such a situation get an additional power supply for the motor and connect the positive terminal to pin 8 of L293D and the ground will go to the common ground.

The working of the project is simple. Once you run the program (explained under software), you will get the output as shown in Fig. 4. Now you enter choices with your keyboard and the motors will move accordingly. Commands in our PHP program includes '+' for rotation in forward direction, '-' for reverse, and 'o' for making the motor oscillate. The angle of motion can also be selected

as shown in Fig. 4. Usually most stepper motors make 200 steps per revolution. Therefore most of the angles are comfortably covered by the small steps.

Software

With PHP and PHP-GPIO libraries already installed, you just have to download the source code (finestep.php) and copy it in '/var/www' directory. Once copied, you can directly run it using below-mentioned command:

```
sudo php -f finestep.php
```

On running the program you will see the output as shown in Fig. 4. Now enter the options and see the motor moving accordingly. ●

The author is an avid user of open source softwares. Professionally, he is a thermal power expert and works as additional general manager at NTPC



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